

## BigchainDB: A scalable blockchain database

BigchainDB offers a scalable, secure, and decentralised data management solution leveraging blockchain technology. It is designed to handle large data volumes with the same level of security and integrity associated with blockchain technology.

BigchainDB's uniqueness lies in its decentralised, tamper-proof database that scales for enterprise applications. Unlike traditional blockchain networks, BigchainDB is a distributed database with high throughput and low latency. It offers features like asset issuance, digital rights management, and decentralised identity management.

### Key features

- BigchainDB can handle thousands of transactions per second, making it suitable for applications requiring high throughput and low latency.
- Operates on a decentralised network of nodes, providing resilience against single points of failure and censorship.
- Once data is written to the BigchainDB, it cannot be altered or deleted, ensuring data integrity and auditability.

- BigchainDB supports a variety of data types and asset types, allowing for the creation of complex data structures and smart assets.

### Limitations

- Operating a BigchainDB network requires significant computational resources and network bandwidth, which may limit its adoption in resource-constrained environments.
- Decentralised networks like BigchainDB may face governance challenges related to consensus mechanisms, network upgrades, and protocol changes.

## MicroPython: Bringing Python to microcontrollers

MicroPython is a lightweight implementation of Python 3 designed for microcontrollers and embedded systems. It enables developers to write Python code directly on microcontroller-based platforms, bringing the simplicity and versatility of Python programming to embedded systems.

MicroPython's uniqueness lies in its ability to simplify embedded development by allowing developers to use Python, a high-level and easy-to-

learn language, instead of low-level languages like C or assembly. This accessibility makes MicroPython suitable for a wide range of developers, including hobbyists, students, and professionals.

CircuitPython, a variant of MicroPython developed by Adafruit Industries, focuses on providing an even more beginner-friendly and educational experience for electronics and programming education.

### Key features

- *Memory efficiency:* MicroPython is designed to operate efficiently in resource-constrained environments. It utilises memory efficiently, enabling it to run on microcontrollers with limited RAM and flash memory. Its memory management system allows for effective utilisation of available resources, making it suitable for embedded systems.
- *Portability and cross-platform compatibility:* It is highly portable and compatible with a wide range of microcontroller architectures and development boards.
- *Interactive development and REPL:* MicroPython provides



Figure 1: Python libraries for developers in various domains